

Raspberry Pi Setup

Saturday, March 01, 2025 1:40 PM

Materials:

- Raspberry Pi
- SD with image already flashed (see How to Install Raspberry Pi OS)
- HDMI adapter cable
- Wired keyboard
- Mouse
- Raspberry Pi power cord

1). Once the SD is ready and in its proper place in the pi (there's a slot on the back for it), plug the power cord into the Pi and the WALL (don't plug into the side of your computer or something. It will not end well for the computer.) Give the Pi a couple of minutes to boot.

- If the light on the bottom left corner of the Pi is red, it has power, but has not yet fully booted.
- Once the Pi has fully booted, the light will turn green. This means you're good to go.

2). Unplug the Pi, then plug it back in. It's not the most graceful way to restart the system, but it works. The light should be a solid green.

3). Plug in a keyboard and mouse to the USB ports on the Pi, and use the red HDMI adapter cable to plug into a monitor. We used the stuff on the desk in the corner of the Makerspace. Make sure you plug back in anything that you unplugged once done.

4). You should be greeted by the Raspberry Pi GUI once everything is set up. Yay! You now have access to the Pi!

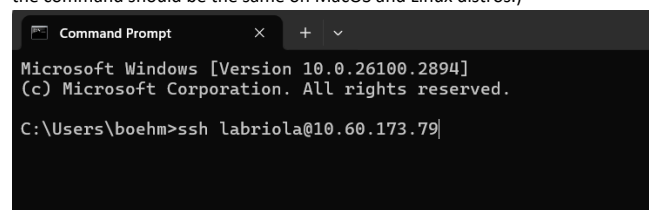
5). Most projects require the Pi to be able to access the Internet. We'll be using a WiFi connection for this. Follow these steps:

- Go to the upper-right corner of the Pi's GUI and open the WiFi options. Select "Mines-Legacy."
- Got to the upper-left corner of the Pi GUI and open a Terminal.
- Run the command "ip address" (note that the quotation marks are NOT part of the command).
- A LOT of information is going to come up. In the section labeled "3: wlan0" there should be a line that says link/ether. The number immediately after this is the WiFi MAC address of the Pi. It should be 12 digits of hexadecimal, split up into pairs by semicolons. For example, the MAC address of our test Pi was 2c:cf:67:0e:12:39. MAKE SURE YOU GRAB THE WLAN0 ADDRESS, NOT THE ETH0 ONE!!!
- Go to netreg.mines.edu on a separate device, ie your laptop. You won't be able to do this step on the Pi since you don't have internet yet on the Pi.
- Enter your correct personal information. In the place where it asks for a MAC address, input the one that you just got from your Pi. If a MAC address auto-filled when you pulled up the form, make sure you overwrite it with the one you actually got off the Pi's command line.
- Register your Pi onto the Mines network. You should get a success message if every went right.
- If you don't have Internet access after 5 minutes, restart your Pi (you'll typically have to restart it). If you still don't have Internet, double check that you MAC address is correct.
- You should now be on the Mines WiFi and able to access the Internet!

6). Now we're able to ssh onto the Pi from our personal devices. To access the Pi's command line remotely, follow these steps:

- In the Pi's GUI, type the command "hostname -I" (Note that the flag is an uppercase i, not l).
- This should show you the IP address of the Pi. Take note that unless we change it, this IP address is NOT static; this means that the IP address will be different every time we turn on the Pi.
- On your laptop, open a command line.
- Type the command "ssh <username>@ip.add.re.ss"
- The username is whatever you put in as the username when you ran the Raspberry Pi Imager (see How to Install Raspberry Pi OS). Additionally, it's also the first part of the username on the command line of your Pi. (Ex: labriola is the username for labriola@labriolapi). The IP address is the one that you just found.

Ex: The Pi username is "labriola" and the IP address is 10.60.173.79. (This is on a Windows machine, but the command should be the same on MacOS and Linux distros.)



```
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\boehm>ssh labriola@10.60.173.79
```



- You will be prompted to authenticate the validity of the host. Type "yes"

```
Command Prompt - ssh labriola x + v
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\boehm>ssh labriola@10.60.173.79
The authenticity of host '10.60.173.79 (10.60.173.79)' can't be established.
ED25519 key fingerprint is SHA256:i+MkeA3QFG89P8uSDMFRoapALiifOdi9tLBzL5OR4mk.
This host key is known by the following other names/addresses:
C:\Users\boehm/.ssh/known_hosts:7: 10.60.0.171
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

- You will then be prompted for a password. Use the one that you created when flashing the Raspberry Pi image initially.

```
labriola@labriolapi: ~ x + v
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\boehm>ssh labriola@10.60.173.79
The authenticity of host '10.60.173.79 (10.60.173.79)' can't be established.
ED25519 key fingerprint is SHA256:i+MkeA3QFG89P8uSDMFRoapALiifOdi9tLBzL5OR4mk.
This host key is known by the following other names/addresses:
C:\Users\boehm/.ssh/known_hosts:7: 10.60.0.171
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.60.173.79' (ED25519) to the list of known hosts.
labriola@10.60.173.79's password:
Linux labriolapi 6.6.51+rpt-rpi-2712 #1 SMP PREEMPT Debian 1:6.6.51-1+rpt3 (2024-10-08) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Nov 19 07:20:36 2024
labriola@labriolapi:~$
```

- Upon success, you should see your Pi's command line. Yay! You've remotely accessed your Pi!

7). If you power off your Pi, the IP address will change, and you will no longer be able to access it with the IP address you had previously. There are two ways to get around this:

- 1: Use nmap (or a similar network scanning tool) to find the new IP address.
- Find the IP address of your personal machine. For example, one Windows you can run the command "ipconfig" on the command line. Then, change the last section of the IP address to be 0/24 instead (ex: 10.60.173.0/24).
 - Run the command "nmap -sP ip.add.re.ss"
 - The Pi should show up under the name "Raspberry Pi Trading." You can double check the MAC address to make sure it's the right one.
 - Unfortunately, this can get tricky because the Mines network is so large. You may have to scan a larger number of IP addresses to find the Pi.
- 2: Change the IP address to be static.
- [How to Set a Static IP Address on Raspberry Pi | Tom's Hardware](#)
 - (I don't think this is actually possible on the Mines network because there might be rules on the router)

Note: On the pis that I've set up recently on the Mines-Legacy network, 2/3 of them have kept the same IP address when I turn them on and off. I'm not sure why this is or if it will continue to happen. Additionally, currently and for the foreseeable future, the Labriola building has not yet been upgraded to the new Mines network.

